Accelerated Algebra and Analysis: Algebra Part

Angus

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Abstract

This is a simple abstract for the Accelerated Algebra and Analysis notes it is focused on the algebra part.

1 Lecture 1

Vectors, Matrices, and Complex numbers

1.1 Geometric motivation

Cubic curves are plain curves given by

$$f(x, y) = 0 \quad \text{where } f \text{ is a cubic polynomial in } x \text{ and } y$$

Theorem 1.1

Chasles's theorem

Let C_3 and C_3' be two cubic curves such that their intersection $C_3 \cap C_3'$ consists of 9 points. Let Z_3 be the set of 8 intersection points

$$Z_3 = (C_3 \cap C_3') \setminus \{P\}$$

where P is the ninth intersection point. Then any cubic curve passing through the points of Z_3 also passes through P.

Chasles's theorem implies Pascal's theorem.

Theorem 1.2

Pascal's theorem

Let A, B, C, D, E, F be the six vertices of a hexagon inscribed in a conic. Then the points of intersection of the pairs of opposite sides $(AB \cap DE), (BC \cap EF), (CD \cap FA)$ are collinear.

- Hexagrammum Mysticum.
- Pappus line

1.2 Vectors

1.2.1 Multidimensional vectors

An n-dimensional vector is an ordered list with n components. It can be represented as

$$\mathbf{v} = (v_1, v_2, \ldots, v_n)$$

We usually write this vector as a column vector

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

Notation

In handwritten notes, an n-dimensional vector $\mathbf{x}$ is often denoted by

$$\overline{x} \quad \underline{x} \quad \mathbf{x} \quad \vec{x}$$

Notation

The set of all n-dimensional vectors is denoted by $\mathbb{R}^n$

$$\mathbb{R}^n = \{(v_1, v_2, \ldots, v_n) \mid v_i \in \mathbb{R} \text{ for } i = 1, 2, \ldots, n\}$$

A special vector in $\mathbb{R}^n$ is the zero vector

$$\mathbf{0} = (0, 0, \ldots, 0)$$

1.2.2 Operations with vectors

Let $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_n)$ be two vectors in $\mathbb{R}^n$.

- **Vector addition:**

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

- **Scalar multiplication:**

$$c\mathbf{x} = \begin{pmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{pmatrix} \quad \text{for } c \in \mathbb{R}$$

Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$. The basic properties of vector operations are

1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$
2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$

3. $c(\mathbf{x}+\mathbf{y}) = c\mathbf{x} + c\mathbf{y}$ for $c \in \mathbb{R}$
4. $(cd)\mathbf{x} = c(d\mathbf{x})$ for $c, d \in \mathbb{R}$
5. $\mathbf{x} + \mathbf{0} = \mathbf{x}$
6. $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$
7. $1\mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$
8. $0\mathbf{x} = \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^n$

1.2.3 Length and distance

Definition 1.1

The length (or norm) of a vector $\mathbf{x} \in \mathbb{R}^n$ is defined as

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

Definition 1.2

The distance between two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined as

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then the distance between $\mathbf{x}$ and $\mathbf{y}$ satisfies the following properties

1. $d(\mathbf{x}, \mathbf{y}) \geq 0$ and $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$
2. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$
3. $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$ (triangle inequality)
4. $d(a\mathbf{x}, a\mathbf{y}) = |a| d(\mathbf{x}, \mathbf{y})$ for $a \in \mathbb{R}$

1.2.4 Dot product

Definition 1.3

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^3$. The dot product of $\mathbf{x}$ and $\mathbf{y}$ is defined by

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2 + x_3 y_3$$

Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ and $c \in \mathbb{R}$. The dot product satisfies the following properties

1. $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$ (commutativity)
2. $(\mathbf{x} + \mathbf{y}) \cdot \mathbf{z} = \mathbf{x} \cdot \mathbf{z} + \mathbf{y} \cdot \mathbf{z}$ (distributivity)
3. $(c\mathbf{x}) \cdot \mathbf{y} = c(\mathbf{x} \cdot \mathbf{y})$ (scalar multiplication)
4. $\mathbf{x} \cdot \mathbf{x} = \|\mathbf{x}\|^2$
5. $\mathbf{x} \cdot \mathbf{0} = 0$

- If $\mathbf{x} \cdot \mathbf{y} = 0$, we call $\mathbf{x}$ and $\mathbf{y}$ orthogonal
- If $|\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\|\|\mathbf{y}\|$, they are collinear

Since

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\| \cos\theta$$

1.2.5 Cauchy-Schwarz inequality

Theorem 1.3

Cauchy-Schwarz inequality

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. The Cauchy-Schwarz inequality states that

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\|\|\mathbf{y}\|$$

with equality if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

Proof. Consider the function

$$f(\lambda) = \|\mathbf{x} + \lambda\mathbf{y}\|^2$$

Since the norm is always non-negative, we have $f(\lambda) \geq 0$ for all $\lambda \in \mathbb{R}$.

$$\|\mathbf{x} + \lambda\mathbf{y}\|^2 = (\mathbf{x} + \lambda\mathbf{y}) \cdot (\mathbf{x} + \lambda\mathbf{y}) = \mathbf{x} \cdot \mathbf{x} + 2\lambda(\mathbf{x} \cdot \mathbf{y}) + \lambda^2(\mathbf{y} \cdot \mathbf{y}) \geq 0$$

This is a quadratic inequality in λ.
For it to hold for all $\lambda \in \mathbb{R}$, the discriminant must be non-positive:

$$(2(\mathbf{x} \cdot \mathbf{y}))^2 - 4(\mathbf{y} \cdot \mathbf{y})(\mathbf{x} \cdot \mathbf{x}) \leq 0$$

Simplifying, we get

$$(\mathbf{x} \cdot \mathbf{y})^2 \leq (\mathbf{x} \cdot \mathbf{x})(\mathbf{y} \cdot \mathbf{y}) = \|\mathbf{x}\|^2\|\mathbf{y}\|^2$$

which is the Cauchy-Schwarz inequality. □

1.2.6 Triangular inequality

Theorem 1.4

Triangular inequality

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. The triangular inequality states that

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$

with equality if and only if one vector is a non-negative scalar multiple of the other.

Proof. Consider the square of the norm of the sum:

$$\|\mathbf{x} + \mathbf{y}\|^2 = (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2$$

Using the Cauchy-Schwarz inequality, we have

$$2(\mathbf{x} \cdot \mathbf{y}) \leq 2\|\mathbf{x}\|\|\mathbf{y}\|$$

Therefore

$$\|\mathbf{x} + \mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2$$

Taking the square root of both sides gives the triangular inequality

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$

□

1.2.7 Angles between vectors

Definition 1.4

The angle θ between two non-zero vectors $\mathbf{x}$ and $\mathbf{y}$ is defined by the formula

$$\cos\theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}$$

Lemma 1.5

Pythagoras theorem for vectors states that if two vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal, then

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$$

If $\mathbf{x}$ and $\mathbf{y}$ are orthogonal (perpendicular), then

$$\mathbf{x} \cdot \mathbf{y} = 0$$

and

$$\theta = \frac{\pi}{2}$$

1.2.8 Cross product of three-dimensional vectors

Definition 1.5

Let $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$ be vectors in $\mathbb{R}^3$.
The cross product $\mathbf{x} \times \mathbf{y}$ is defined as

$$\mathbf{x} \times \mathbf{y} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}$$

where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard unit vectors in $\mathbb{R}^3$.

$$\mathbf{x} \times \mathbf{y} = \begin{pmatrix} x_2y_3 - x_3y_2 \\ -x_1y_3 + x_3y_1 \\ x_1y_2 - x_2y_1 \end{pmatrix}$$

The cross product has the following properties:

1. $\mathbf{x} \times \mathbf{y} = -(\mathbf{y} \times \mathbf{x})$
2. $\mathbf{x} \times (\mathbf{y} + \mathbf{z}) = \mathbf{x} \times \mathbf{y} + \mathbf{x} \times \mathbf{z}$
3. $(a\mathbf{x}) \times \mathbf{y} = a(\mathbf{x} \times \mathbf{y})$ for $a \in \mathbb{R}$
4. $\|\mathbf{x} \times \mathbf{x}\| = 0$
5. $(\mathbf{x} \times \mathbf{y}) \cdot \mathbf{x} = 0$ and $(\mathbf{x} \times \mathbf{y}) \cdot \mathbf{y} = 0$
6. $\|\mathbf{x} \times \mathbf{y}\| = \|\mathbf{x}\|\|\mathbf{y}\| \sin\theta$, where θ is the angle between $\mathbf{x}$ and $\mathbf{y}$

1.3 Matrix

1.3.1 Definition

Definition 1.6

A matrix is a rectangular array of numbers arranged in rows and columns. An $m \times n$ matrix has m rows and n columns and is typically denoted by

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

Zero matrix is a matrix in which all the elements are zero, typically denoted by

$$0 = \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}$$

$n \times n$ identity matrix is a matrix in which all the elements on the main diagonal are 1 and all other elements are 0, typically denoted by

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

1.3.2 Sum representation

Any $m \times n$ matrix $A = (a_{ij})$ can be represented as the sum of its elements multiplied by the corresponding standard basis matrices

$$A = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} E_{ij},$$

where E_{ij} is the $m \times n$ matrix with 1 in the (i, j)-th position and 0 elsewhere.

1.3.3 Algebra operations with matrices

$A + B$

$$(A + B)_{ij} = A_{ij} + B_{ij}$$

cA

$$(cA)_{ij} = c \cdot A_{ij}$$

AB

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

1.3.4 Matrix multiplication

Definition 1.7

matrix multiplication

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \quad B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1q} \\ b_{21} & b_{22} & \dots & b_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ b_{p1} & b_{p2} & \dots & b_{pq} \end{pmatrix}$$

$$AB = \begin{pmatrix} (AB)_{11} & (AB)_{12} & \dots & (AB)_{1q} \\ (AB)_{21} & (AB)_{22} & \dots & (AB)_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ (AB)_{m1} & (AB)_{m2} & \dots & (AB)_{mq} \end{pmatrix}$$

Make sure that the number of columns of A matches the number of rows of B for the matrix multiplication AB to be defined,

Example

If A is an $m \times n$ matrix and B is a $p \times q$ matrix, then $n = p$. In particular, the product AB will be an $m \times q$ matrix.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}_{m \times n}, \quad B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1q} \\ b_{21} & b_{22} & \dots & b_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ b_{p1} & b_{p2} & \dots & b_{pq} \end{pmatrix}_{p \times q}$$

$$AB = \begin{pmatrix} (AB)_{11} & (AB)_{12} & \dots & (AB)_{1q} \\ (AB)_{21} & (AB)_{22} & \dots & (AB)_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ (AB)_{m1} & (AB)_{m2} & \dots & (AB)_{mq} \end{pmatrix}_{m \times q}$$

Basic properties of matrix addition and scalar multiplication

1. $(A + B) = (B + A)$

2. $(A+B)+C = A+(B+C)$
3. $A+0 = A$
4. $c(A+B) = cA+cB, \quad c \in \mathbb{R}$

Basic properties of matrix multiplication

1. $(AB)C = A(BC)$
2. $A(B+C) = AB+AC$
3. $(A+B)C = AC+BC$
4. $AI_n = A = I_m A_{m\times n}$

For $(AB)C = A(BC)$

Proof. Let A be an $m \times n$ matrix, B be an $n \times p$ matrix, and C be a $p \times q$ matrix. Then

$$\begin{aligned}((AB)C)_{ij} &= \sum_{k=1}^{p}(AB)_{ik}C_{kj} \\ &= \sum_{k=1}^{p}\left(\sum_{l=1}^{n}A_{il}B_{lk}\right)C_{kj} \\ &= \sum_{l=1}^{n}A_{il}\left(\sum_{k=1}^{p}B_{lk}C_{kj}\right) \\ &= (A(BC))_{ij}, \quad 1 \le i \le m, \quad 1 \le j \le q\end{aligned}$$

Therefore, $(AB)C = A(BC)$. □

1.3.5 Column Vectors

Definition 1.8

Column vectors $(x_1, x_2, \ldots, x_m)^T$

A column vector is an $m \times 1$ matrix, i.e.,

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix}_{m\times 1}$$

If

$$A = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix}$$

Then

$$MA = \begin{pmatrix} (MA)_1 \\ (MA)_2 \\ \vdots \\ (MA)_m \end{pmatrix}$$

1.3.6 Invertible matrices

Definition 1.9

An $n \times n$ matrix A is said to be invertible if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n.$$

The matrix B is called the inverse of A and is denoted by A^{-1}.

A matrix is called singular if it is an $n \times n$ matrix that is not invertible.

Notation

The set of all invertible $n \times n$ matrices is denoted by $GL(n, \mathbb{R})$ if the entries are real numbers.

For 2×2 case, an invertible matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has an inverse given by

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad ad - bc \neq 0.$$

1.3.7 Transpose

Definition 1.10

Let A be an $m \times n$ matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}_{m \times n}.$$

The transpose of A, denoted by A^T, is the $n \times m$ matrix obtained by interchanging the rows and columns of A:

$$A^T = \begin{pmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{pmatrix}_{n \times m}.$$

Properties of the transpose

1. $(A^T)^T = A$

2. $(A+B)^T = A^T + B^T$
3. $(cA)^T = cA^T$ for any scalar c
4. $(AB)^T = B^T A^T$

Definition 1.11

Symmetric matrix A matrix A is called symmetric if $A^T = A$.

Skew-symmetric matrix A matrix A is called skew-symmetric if $A^T = -A$.

Theorem 1.6

For any two $n \times n$ matrices A and B, the matrix $AB - BA$ is skew-symmetric.

Proof. We need to show that $(AB - BA)^T = -(AB - BA)$.

$$\begin{aligned}(AB - BA)^T &= (AB)^T - (BA)^T \\ &= B^T A^T - A^T B^T \\ &= -(A^T B^T - B^T A^T) \\ &= -(AB - BA).\end{aligned}$$

Hence, $AB - BA$ is skew-symmetric. □

1.4 Complex Number

1.4.1 Definition

We adjoin the imaginary unit i in $\mathbb{R}$, where

$$i^2 = -1$$

We define a complex number as

$$z = a + bi, \quad a, b \in \mathbb{R}.$$

And we denote the set of all complex numbers by $\mathbb{C}$

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$$

1.4.2 Fundamental theorem of Algebra

Theorem 1.7

Let

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

be a non-constant polynomial with complex coefficients.
If $n \geq 1$, then there exists at least one complex number $z_0 \in \mathbb{C}$ such that

$$p(z_0) = 0.$$

In other words, every non-constant polynomial with complex coefficients has at least one root in $\mathbb{C}$.

Lemma 1.8

Let

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

be a non-constant polynomial with complex coefficients.
Then $p(z)$ can be factored into linear factors over $\mathbb{C}$, i.e.,

$$p(z) = a_n (z - z_1)(z - z_2) \dots (z - z_n),$$

where $z_1, z_2, \dots, z_n \in \mathbb{C}$ are the roots of $p(z)$.

1.5 Field

1.5.1 Definition

Definition 1.12

A field is a set F equipped with addition and multiplication, such that the following properties hold

1. F is an abelian group under addition, with identity element 0.
2. $F \setminus \{0\}$ is an abelian group under multiplication, with identity element 1.
3. Multiplication is distributive over addition, i.e., for all $a, b, c \in F$,

$$a(b + c) = ab + ac.$$

Definition 1.13

Abelian group

A group G satisfies the commutative property if for all $a, b \in G$

$$a + b = b + a$$

1.5.2 Other finite fields

To construct many finite fields, fix a prime number $p \geq 2$. Let $\mathbb{F}_p$ be the following set consisting of p integers:

$$\{0, 1, \dots, p - 1\}.$$

Equip $\mathbb{F}_p$ with operations $+$ and $*$ as follows:

1. For every two elements a and b in $\mathbb{F}_p$, let $a + b =$ the remainder of the integer $a + b$ when divided by p.
2. For every two elements a and b in $\mathbb{F}_p$, let $a * b =$ the remainder of the integer $a * b$ when divided by p.

Then $\mathbb{F}_p$ equipped with $+$ and $*$ is a field.

Example

Suppose that $p = 1973$ (it is prime). Then

$$\frac{a}{45} = a * 570$$

for every a in $\mathbb{F}_p$, since $45 * 570 = 1$ in $\mathbb{F}_p$.

1.5.3 Solving system of two linear equations

Example

Let us solve the following system of equations in $\mathbb{F}_7$:

$$\begin{cases} 2x + 3y = 5, \\ 3x + 4y = 1. \end{cases}$$

Observe that $2 \times 4 = 1$ in $\mathbb{F}_7$. Then $\frac{1}{2} = 4$ and

$$2x + 3y = 5 \implies x = 4(5 - 3y),$$

which gives $x = 6 + 2y$. Then

$$3(6 + 2y) + 4y = 1,$$

so that $3y = 4$, which gives $y = 5 \times 4 = 6$. Then

$$(x, y) = (6 + 2 \times 6, 6) = (4, 6).$$

1.5.4 A better way to construct a field with 4 elements

Start with the field $\mathbb{F}_2$ consisting of two elements 0 and 1.
Introduce a new element t such that

$$t^2 = t + 1$$

Consider all sums (there are 4 of them) that look like

$$a + bt$$

where a and b are both in $\mathbb{F}_2$. Denote the obtained set by $\mathbb{F}_4$. Then extend $+$ and $*$ from $\mathbb{F}_2$ to elements in $\mathbb{F}_4$ as follows:

$$(a + bt) * (c + dt) = ac + bdt^2 + adt + bct = (ac + bd) + (ad + bc + bd)t$$

and

$$(a + bt) + (c + dt) = (a + c) + (b + d)t$$

for any elements a, b, c, d in the field $\mathbb{F}_2$. Then $\mathbb{F}_4 = \{0, 1, t, 1 + t\}$ is a field.

2 Lecture 2

Linear subspaces and their bases

2.1 Linear Subspaces

2.1.1 Hyperplanes

Definition 2.1

A hyperplane in $\mathbb{R}^n$ is a set of points x in $\mathbb{R}^n$ that satisfy a linear equation of the form

$$\sum_{i=1}^{n} A_i x_i = B,$$

where $A_1, A_2, \ldots, A_n$ are not all zero, and $A_1, \ldots, A_n, B$ are in $\mathbb{R}$.

Remark

A hyperplane contains the origin if and only if $B = 0$.

2.1.2 Linear subspaces

Definition 2.2

A linear subspace (or vector subspace) of $\mathbb{R}^n$ is a set of points x in $\mathbb{R}^n$ such that

- if x and y are in the subspace V, then

$$x + y \in V.$$

- if x is in the subspace and α is a scalar in $\mathbb{R}$, then

$$\alpha x \in V.$$

- the zero vector 0 is in the subspace.

$$0 \in V.$$

Notation

Linear subspace = Vector subspace

Example

- $\{0\}$ is a linear subspace of $\mathbb{R}^n$, which is trivial linear subspace.
- a line through the origin in $\mathbb{R}^n$ is also a linear subspace.
- a hyperplane through the origin in $\mathbb{R}^n$ is also a linear subspace.
- whole $\mathbb{R}^n$ is also a linear subspace.

2.1.3 Kernel

Definition 2.3

Kernel of a matrix A is the set of all vectors x such that $Ax = 0$.

$$\ker(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}$$

the kernel is a linear subspace of $\mathbb{R}^n$.

Notation

Some people called the kernel of a matrix A the null space of A.

Proof. To prove that the kernel is a linear subspace, we need to show that it satisfies the three conditions for a linear subspace

- If x and y are in $\ker(A)$, then $Ax = 0$ and $Ay = 0$.

  $$A(x + y) = Ax + Ay = 0 + 0 = 0,$$

  which means $x + y \in \ker(A)$.

- If x is in $\ker(A)$ and α is a scalar in $\mathbb{R}$, then $Ax = 0$.

  $$A(\alpha x) = \alpha Ax = \alpha \cdot 0 = 0,$$

  which means $\alpha x \in \ker(A)$.

- The zero vector 0 is in $\ker(A)$ because

  $$A0 = 0.$$

□

2.2 Linear Combinations and Span

2.2.1 Linear Combination

Definition 2.4

A linear combination of vectors $v_1, v_2, \ldots, v_k$ in $\mathbb{R}^n$ is a vector of the form

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k,$$

where $\alpha_1, \alpha_2, \ldots, \alpha_k$ are scalars in $\mathbb{R}$.

Example

Matrix multiplication can be seen as a linear combination of the columns of the matrix. For example, if A is an $m \times n$ matrix with columns $A_1, A_2, \ldots, A_n$ and x is a vector in $\mathbb{R}^n$ with components $x_1, x_2, \ldots, x_n$, then

$$Ax = x_1 A_1 + x_2 A_2 + \cdots + x_n A_n.$$

Let V contains list of vectors $v_1, v_2, \ldots, v_k$ in $\mathbb{R}^n$. Then it contains all linear combinations of these vectors, i.e.,

$$\{\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k \mid \alpha_1, \alpha_2, \ldots, \alpha_k \in \mathbb{R}\} \subseteq V.$$

Remark

The zero is a linear combination for all sets of vectors, obtained by taking all the scalars $\alpha_1, \alpha_2, \ldots, \alpha_k$ to be zero.

2.2.2 Linear Span

Definition 2.5

The linear span of a set of vectors $v_1, v_2, \ldots, v_k$ in $\mathbb{R}^n$ is the set of all linear combinations of these vectors, i.e.,

$$\operatorname{span}\{v_1, v_2, \ldots, v_k\} = \{\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k \mid \alpha_1, \alpha_2, \ldots, \alpha_k \in \mathbb{R}\}.$$

Notation

The linear span of vectors $v_1, v_2, \ldots, v_k$ is denoted by $\operatorname{span}\{v_1, v_2, \ldots, v_k\}$. Or simply by $\langle v_1, v_2, \ldots, v_k \rangle$.

In some sense the linear span is a technique to gerente a set of vectors from some (a list of) vectors.

2.2.3 Spanning Sets

Definition 2.6

A set of vectors $v_1, v_2, \ldots, v_k$ in $\mathbb{R}^n$ is called a spanning set of a vector space V if

$$V = \operatorname{span}\{v_1, v_2, \ldots, v_k\}.$$

Example

- The standard basis vectors $v_1 = (1, -1, 0), v_2 = (0, 1, -1)$ in $\mathbb{R}^3$ form a spanning set of a plane in $\mathbb{R}^3$.
$$x_1 + x_2 + x_3 = 0 \subset \mathbb{R}^3$$
- Let $e_1, e_2, \ldots, e_n$ be the standard basis vectors in $\mathbb{R}^n$. Then $\{e_1, e_2, \ldots, e_n\}$ forms a spanning set of $\mathbb{R}^n$, i.e.,
$$\mathbb{R}^n = \operatorname{span}\{e_1, e_2, \ldots, e_n\}.$$

Remark

If $\{v_1, v_2, \ldots, v_k\}$ are contained in $\operatorname{span}\{u_1, u_2, \ldots, u_m\}$, i.e.,

$$v_i \in \operatorname{span}\{u_1, u_2, \ldots, u_m\} \quad \text{for all } i = 1, 2, \ldots, k,$$

then

$$\operatorname{span}\{v_1, v_2, \ldots, v_k\} \subseteq \operatorname{span}\{u_1, u_2, \ldots, u_m\}.$$

2.2.4 Column Space and Row Space

Let A be an $m \times n$ matrix. Then its column are vectors in $\mathbb{R}^m$.

Definition 2.7

The column space of A, is the linear span of its column vectors.

$$\text{Col}(A) = \{y \in \mathbb{R}^m \mid y = Ax \text{ for some } x \in \mathbb{R}^n\}.$$

Definition 2.8

Let A be an $m \times n$ matrix. The row space of A, is the linear span of its row vectors.

$$\text{Row}(A) = \{y \in \mathbb{R}^n \mid y = xA \text{ for some } x \in \mathbb{R}^m\}.$$

2.3 Linear Independence and Dimension

2.3.1 Linearly Dependence

Definition 2.9

A set of vectors $\{v_1, v_2, \ldots, v_k\}$ in a vector space V is said to be linearly dependent if there exist scalars $\lambda_1, \lambda_2, \ldots, \lambda_k$, not all zero, such that

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_k v_k = 0.$$

If $\lambda_m \neq 0$, then

$$v_m = -\frac{\lambda_1}{\lambda_m} v_1 - \frac{\lambda_2}{\lambda_m} v_2 - \cdots - \frac{\lambda_{m-1}}{\lambda_m} v_{m-1}.$$

2.3.2 Linearly Independence

Definition 2.10

A set of vectors $\{v_1, v_2, \ldots, v_k\}$ in a vector space V is said to be linearly independent if it is not linearly dependent.

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_k v_k = 0,$$

where the only solution is $\lambda_1 = \lambda_2 = \cdots = \lambda_k = 0$.

Example

$e_1, e_2, \ldots, e_n \in \mathbb{R}^n$ are linearly independent. To see this, consider the equation

$$\lambda_1 e_1 + \lambda_2 e_2 + \cdots + \lambda_n e_n = 0.$$

Since $e_1, e_2, \ldots, e_n$ are the standard basis vectors, the only solution is $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$.

Let a,b,c,d be some real numbers, and let

$$v_1 = \begin{pmatrix} a \\ b \end{pmatrix}, \quad v_2 = \begin{pmatrix} c \\ d \end{pmatrix}.$$

Then v_1 and v_2 are linearly independent if and only if the determinant of the matrix formed by placing v_1 and v_2 as columns is non-zero

$$\det \begin{pmatrix} a & c \\ b & d \end{pmatrix} = ad - bc \neq 0.$$

2.3.3 Linear Independence versus Linear Combination

Theorem 2.1

Let $v_1, v_2, \ldots, v_k$ be vectors in $\mathbb{R}^n$. The following statements are equivalent

- $v_1, v_2, \ldots, v_k$ are linearly independent.
- None of the vectors $v_1, v_2, \ldots, v_k$ can be written as a linear combination of the others.
- for every $x \in span\{v_1, v_2, \ldots, v_k\}$, there exists a unique set of scalars $\lambda_1, \lambda_2, \ldots, \lambda_k$ such that
$$x = \sum_{i=1}^{k} \lambda_i v_i.$$

Proof. consider x and x' in $\text{span}\{v_1, v_2, \ldots, v_k\}$ such that

$$x = \sum_{i=1}^{k} \lambda_i v_i \quad \text{and} \quad x' = \sum_{i=1}^{k} \lambda_i' v_i.$$

If $x = x'$, then

$$\sum_{i=1}^{k} \lambda_i v_i = \sum_{i=1}^{k} \lambda_i' v_i \implies \sum_{i=1}^{k} (\lambda_i - \lambda_i') v_i = 0.$$

Since $v_1, v_2, \ldots, v_k$ are linearly independent, it follows that

$$\lambda_i - \lambda_i' = 0$$

for all i, i.e., $\lambda_i = \lambda_i'$ for all i. This proves the uniqueness of the representation. □

Corollary 2.2

Suppose $v_1, v_2, \ldots, v_k$ are linearly independent. Then any sublist of $v_1, v_2, \ldots, v_k$ is also linearly independent.

2.3.4 Steinitz Exchange Lemma

Lemma 2.3

Steinitz exchange lemma

- Let V be a vector subspace of $\mathbb{R}^n$.
- Let $\{v_1, v_2, \ldots, v_m\}$ be a linearly independent set in V and let $v \in V$.
- Let $\{w_1, w_2, \ldots, w_k\}$ be a set spanning V.

One has

$$m \leq k.$$

Corollary 2.4

Any $k \geq n$ vectors in $\mathbb{R}^n$ are linearly dependent.

Example

Let M be an $n \times n$ matrix. There exists a polynomial

$$p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

Such that

$$p(M) = M^n + a_{n-1}M^{n-1} + \cdots + a_1 M + a_0 I = 0.$$

2.3.5 Basis of a linear subspace

Definition 2.11

A basis of a linear subspace V of $\mathbb{R}^n$ is a linearly independent set of vectors in V that spans V.

$$V = \text{span}\{v_1, v_2, \ldots, v_k\}.$$

Theorem 2.5

Any two bases of a finite-dimensional vector space have the same number of elements.

Proof. We have two bases of a finite-dimensional vector space V.

$$\{v_1, v_2, \ldots, v_m\} \text{ and } \{w_1, w_2, \ldots, w_n\}$$

By the Steinitz exchange lemma, we have

$$m \leq n \text{ and } n \leq m.$$

Therefore,

$$m = n.$$

□

Lemma 2.6

Delection Lemma

If $v_1, v_2, \ldots, v_k$ is a linearly dependent set in a vector space V and

$$V \subseteq \text{span}\{v_1, v_2, \ldots, v_k\},$$

We can find an index i such that v_i can be expressed as a linear combination of the other vectors in the set.

$$v_i \in \text{span}\{v_1, v_2, \ldots, v_{i-1}, v_{i+1}, \ldots, v_k\}.$$

v_i is called a deleted vector.
Then we can remove v_i from the set without changing the span of the set.

Lemma 2.7

Extension Lemma

If $v_1, v_2, \ldots, v_k$ is a linearly independent set in a vector space V and

$$\text{span}\{v_1, v_2, \ldots, v_k\} \subseteq V,$$

then there exists an index i such that v_i can be expressed as span of the other vectors in the set.

$$v_i \in \text{span}\{v_1, v_2, \ldots, v_{i-1}, v_{i+1}, \ldots, v_k\}.$$

2.3.6 Dimension

Definition 2.12

The dimension of a vector space V is the number of vectors in any basis of V.

Lemma 2.8

Let W be a linear subspace of $\mathbb{R}^n$ such that $V \subseteq W$. Then

$$\dim(V) \leq \dim(W).$$

Moreover, if $\dim(V) = \dim(W)$, then $V = W$.

2.4 Arbitrary Field $\mathbb{F}$

Let $\mathbb{F}$ be an arbitrary field, Everything we did so far could be done in $\mathbb{F}^n$ instead of $\mathbb{R}^n$.

Theorem 2.9

If $\mathbb{F}$ is a finite field, then

$$|\mathbb{F}| = p^n.$$

for some prime number p and positive integer n.

Proof. Since $\mathbb{F}$ is a finite field

$$1, 1+1, 1+1+1, \ldots$$

Since $\mathbb{F}$ is finite, this sequence must eventually repeat, implying that there exists a positive integer p such that

$$\underbrace{1+1+\cdots+1}_{p \text{ times}} = 0.$$

So since p is the smallest positive integer such that the sum of 1 repeated p times equals 0, we have

$$\mathbb{F}_p \subseteq \mathbb{F}.$$

Then $\mathbb{F}$ can be viewed as a vector space over $\mathbb{F}_p$. Let $n = \dim_{\mathbb{F}_p}(\mathbb{F})$. Then

$$|\mathbb{F}| = p^n.$$

□

2.5 Orthogonality

2.5.1 Orthogonal Vectors

Definition 2.13

Vectors $v_1, v_2, \ldots, v_k$ in a vector space V are called orthogonal if

$$\langle v_i, v_j \rangle = 0 \quad \text{for all } i \neq j.$$

We say that these vectors are orthonormal if

$$v_i \cdot v_j = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Lemma 2.10

If $v_1, v_2, \ldots, v_k$ are orthogonal vectors in a vector space V, then they are linearly independent.

Corollary 2.11

Orthonormal vectors in a vector space V are linearly independent.

2.5.2 Orthonormal Basis

Definition 2.14

Orthonormal basis of a vector space V is a basis consisting of orthonormal vectors.

Let x be a vector in a vector space V with an orthonormal basis $\{v_1, v_2, \ldots, v_n\}$. Then x can be expressed as

$$x = \sum_{i=1}^{n} \langle x, v_i \rangle v_i.$$

Much easier to compute the dot product with the orthonormal basis.

2.5.3 Orthonormal Complement

Definition 2.15

Let V be a vector space with an inner product, and let W be a subspace of V. The orthonormal complement of W in V, denoted by $W^\perp$, is the set of all vectors in V that are orthogonal to every vector in W:

$$W^\perp = \{v \in V \mid \langle v, w\rangle = 0 \text{ for all } w \in W\}.$$

The subset $W^\perp$ is itself a subspace of V.

Lemma 2.12

Let $v_1, v_2, \ldots, v_m$ be vectors in V that span V Then

$$V^\perp = \{v \in V \mid \langle v, v_i\rangle = 0 \text{ for all } i = 1, 2, \ldots, m\}.$$

2.5.4 Orthogonal Decomposition

Theorem 2.13

Let V be a linear subspace of $\mathbb{R}^n$. Then

1. $V \cap V^\perp = \{0\}$
2. for each $x \in \mathbb{R}^n$, one has
$$x = v + w,$$
for unique $v \in V$ and $w \in V^\perp$.
3. $\dim V + \dim V^\perp = n$

Idea of proof: let $v = P(x)$ and $w = x - P(x)$ for

$$P(x) = \sum_{i=1}^{m} \langle x, v_i\rangle v_i.$$

Then $v \in V$ and $w \in V^\perp$, and the decomposition is unique.

2.5.5 Orthogonal Projection

Definition 2.16

Let V be a linear subspace of $\mathbb{R}^n$, and let $x \in \mathbb{R}^n$. The orthogonal projection of x onto V, denoted by $P_V(x)$, is the vector $v \in V$ such that

$$x - v \in V^\perp.$$

If $\{v_1, v_2, \ldots, v_m\}$ is an orthonormal basis of V, then

$$P_V(x) = \sum_{i=1}^{m} \langle x, v_i\rangle v_i.$$

2.5.6 Orthonormal Extension

Lemma 2.14

There is some orthonormal basis of V that contains the given orthonormal set $\{v_1, v_2, \ldots, v_m\}$.

Proof. As $\{v_1, \ldots, v_m\}$ is orthonormal, it is linearly independent. Extend it to a basis of V.

$$\{v_1, \ldots, v_m, u_1, \ldots, u_k\}$$

Applying the Gram-Schmidt process to this basis leaves the first m vectors unchanged, and orthogonalizes the remaining vectors to yield an orthonormal set

$$\{e_{m+1}, \ldots, e_{m+k}\}$$

Therefore

$$\{v_1, \ldots, v_m, e_{m+1}, \ldots, e_{m+k}\}$$

forms an orthonormal basis for V containing the given set. □

2.5.7 The Gram-Schmidt Process

Theorem 2.15

Gram-Schmidt Process

Let $\{u_1, u_2, \ldots, u_m\}$ be a linearly independent set of vectors in $\mathbb{R}^n$.
The *Gram-Schmidt process* constructs an orthonormal set $\{v_1, v_2, \ldots, v_m\}$ as follows

$$w_1 = u_1, \qquad v_1 = \frac{w_1}{\|w_1\|},$$
$$w_j = u_j - \sum_{i=1}^{j-1} \langle u_j, v_i \rangle v_i, \quad v_j = \frac{w_j}{\|w_j\|}, \quad j = 2, \ldots, m.$$

Then $\{v_1, v_2, \ldots, v_m\}$ is an orthonormal set that spans the same subspace as $\{u_1, u_2, \ldots, u_m\}$.

The step of gram-schmidt process involves subtracting the projections of the current vector onto the previously obtained orthonormal vectors and then normalizing the result.

1. $w_1 = u_1$
2. $w_2 = u_2 - \langle u_2, v_1 \rangle v_1$
3. $w_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$
4. ...
5. $w_m = u_m - \sum_{i=1}^{m-1} \langle u_m, v_i \rangle v_i$
6. Then we normalize each w_i to obtain

$$v_i = \frac{w_i}{\|w_i\|} \qquad \text{for } i = 1, 2, \ldots, m$$

3 Lecture 3

Rank theorem, determinants and RREF

3.1 Row Space, Column Space and Kernal

Definition 3.1

- The *row space* of a matrix A is the subspace of $\mathbb{R}^n$ spanned by the rows of A.

$$\text{row}(A) = \text{span}\{\text{rows of } A\} \subseteq \mathbb{R}^n$$

- The *column space* of a matrix A is the subspace of $\mathbb{R}^m$ spanned by the columns of A.

$$\text{col}(A) = \text{span}\{\text{columns of } A\} \subseteq \mathbb{R}^m$$

- The *kernel* (or null space) of a matrix A is the set of all vectors $x \in \mathbb{R}^n$ such that $Ax = 0$.

$$\ker(A) = \{x \in \mathbb{R}^n \mid Ax = 0\} \subseteq \mathbb{R}^n$$

Lemma 3.1

The following inqualities hold

$$\dim(\text{col}(A))) \leq \min(m, n)$$

$$\dim(\text{row}(A)) \leq \min(m, n)$$

3.2 Rank–Nullity Theorem

Lemma 3.2

For any $m \times n$ matrix A,

$$\ker(A) = \text{row}(A)^{\perp},$$

$$\text{row}(A) = \ker(A)^{\perp}.$$

Corollary 3.3

Let A be an $m \times n$ matrix. Then the dimension of the row space plus the dimension of the kernel equals the number of columns of A.

$$\dim(\text{row}(A)) + \dim(\ker(A)) = n.$$

Theorem 3.4

Rank–Nullity Theorem

For any $m \times n$ matrix A,

$$\operatorname{rank}(A) + \dim(\ker(A)) = n.$$

where $\operatorname{rank}(A) = \dim(\operatorname{col}(A))$ and $\operatorname{nullity}(A) = \dim(\ker(A))$.

Remark

Therefore we have

$$\operatorname{rank}(A) = \dim(\operatorname{col}(A)) = \dim(\operatorname{row}(A)).$$

Proof. let

$$\begin{aligned} w_1, w_2, \ldots, w_l &\in \mathbb{R}^n \quad \text{be a basis of } \operatorname{col}(A), \\ v_1, v_2, \ldots, v_k &\in \mathbb{R}^n \quad \text{be a basis of } \ker(A). \end{aligned}$$

We need to show that the set $\{w_1, w_2, \ldots, w_l, v_1, v_2, \ldots, v_k\}$ forms a basis of $\mathbb{R}^n$ i.e.,

$$l + k = n.$$

We can choose $v_{k+1}, v_{k+2}, \ldots, v_n \in \mathbb{R}^n$ such that

$$Av_{k+i} = w_i, \quad i = 1, 2, \ldots, l.$$

We claim that $v_1, v_2, \ldots, v_{k+l}$ span $\mathbb{R}^n$.

$$Ax = \sum_{i=1}^{l} \alpha_i w_i, \quad \alpha_i \in \mathbb{R}.$$

For some $\alpha_1, \alpha_2, \ldots, \alpha_l \in \mathbb{R}$, we have

$$Ax = \sum_{i=1}^{l} \alpha_i w_i.$$

Let

$$y = x - \sum_{i=1}^{l} \alpha_i v_{k+i}.$$

Then

$$Ay = Ax - \sum_{i=1}^{l} \alpha_i A v_{k+i} = Ax - \sum_{i=1}^{l} \alpha_i w_i = 0,$$

so $y \in \ker(A)$. Hence we can write

$$y = \sum_{i=1}^{k} \beta_i v_i.$$

Therefore,

$$x = y + \sum_{i=1}^{l} \alpha_i v_{k+i} = \sum_{i=1}^{k} \beta_i v_i + \sum_{i=1}^{l} \alpha_i v_{k+i},$$

which shows that x is a linear combination of $v_1, v_2, \ldots, v_{k+l}$, and hence

$$\text{span}\{v_1, v_2, \ldots, v_{k+l}\} = \mathbb{R}^n.$$

To complete the proof that they form a basis of $\mathbb{R}^n$, we need to show that they are linearly independent, suppose

$$\sum_{i=1}^{k} \gamma_i v_i + \sum_{i=1}^{l} \delta_i v_{k+i} = 0.$$

Applying A to both sides gives

$$\sum_{i=1}^{l} \delta_i w_i = 0.$$

Since $w_1, w_2, \ldots, w_l$ are linearly independent, we have $\delta_i = 0$ for all i. Then

$$\sum_{i=1}^{k} \gamma_i v_i = 0.$$

Since $v_1, v_2, \ldots, v_k$ are linearly independent, we have

$$\gamma_1 = \gamma_2 = \cdots = \gamma_k = 0.$$

Hence $v_1, v_2, \ldots, v_{k+l}$ are linearly independent. □

3.3 Rank of a Matrix

Definition 3.2

The *rank* of a matrix $A \in \mathbb{R}^{m \times n}$ is

$$\text{rank}(A) = \dim(\text{col}(A)) = \dim(\text{row}(A)).$$

Lemma 3.5

If A and B are $n \times n$ matrices such that

$$AB = I_n \implies BA = I_n.$$

Proof.

$$AB = I_n \implies BA = I_n.$$

Let

$$C = BA.$$

Suppose $C \neq I_n$. Then

$$AC = A(BA) = (AB)A = I_n A = A.$$

Since A is invertible, we can multiply both sides by A^{-1} to get

$$C - I_n = 0.$$

Which gives

$$\ker(A) \neq \{0\}.$$

This contradicts the assumption that A is invertible, since an invertible matrix has a trivial kernel. □

Theorem 3.6

The following are equivalent

1. the matrix A is invertible,
2. There exists a $n \times n$ matrix B such that $AB = BA = I_n$,
3. $\ker(A) = \{0\}$,
4. ...

3.4 Chasles's Theorem

Definition 3.3

A *cubic curve* is a curve in the plane defined by a polynomial equation of degree 3 in two variables.

$$a_1x^3 + a_2x^2y + a_3xy^2 + a_4y^3 + a_5x^2 + a_6xy + a_7y^2 + a_8x + a_9y + a_{10} = 0$$

Where $a_1, a_2, \ldots, a_{10}$ are constants.

Theorem 3.7

Let C_1, C_2 be cubic curves such that the interaction consists 9 points.

$$C_1 \cap C_2 = \{P_1, P_2, \ldots, P_9\}.$$

Let C be a cubic curve passing through eight of the nine points $P_1, P_2, \ldots, P_8$. Then C must also pass through the ninth point P_9.

Remark

This theorem implies the addition of points on a cubic curve is associative.

3.5 Determinants

3.5.1 Examples

2×2 determinants are defined as follows

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

3×3 determinants are defined as follows

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \det \begin{pmatrix} d & f \\ g & i \end{pmatrix} + c \det \begin{pmatrix} d & e \\ g & h \end{pmatrix}$$

3.5.2 Inductive Approach

The determinant of an $n \times n$ matrix $A = (a_{ij})$ can be defined inductively as follows:

- For $n = 1$, $\det(a_{11}) = a_{11}$.
- For $n > 1$,

$$\det(A) = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det(A_{1j}),$$

where A_{1j} is the $(n-1) \times (n-1)$ matrix obtained by removing the first row and the j-th column of A.

Remark

It can be use in any column and any row for the expansion.

$$\det(A) = \sum_{j=1}^{n} (-1)^{1+j} a_{ij} \det(A_{ij}), \quad i = 1, 2, \ldots, n$$

or

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+1} a_{ij} \det(A_{ij}). \quad j = 1, 2, \ldots, n$$

3.5.3 Properties

The following properties hold for the determinant of a square matrix.

Lemma 3.8

- Swapping two rows or two columns of a matrix multiplies its determinant by -1.

$$\det A' = -\det A$$

- Multiplying a row or a column by a scalar k multiplies the determinant by k.

$$\det A' = k \det A$$

- if a matrix has a row or column of zeros, its determinant is zero.

$$\det A = 0$$

- Let B be a matrix with columns $u_1, u_2, \ldots, u_n$, let C be a matrix with columns $v_1, v_2, \ldots, v_n$. If exists k such that $w_k = u_i + v_i$ and $w_i = v_i = u_i$ for all $i \neq k$, then

$$\det(C) = \det(A) + \det(B).$$

- $\det(AB) = \det(A) \det(B)$ for any square matrices A and B.
- $\det(A^T) = \det(A)$ for any square matrix A.

And we have some further properties for the determinant of a square matrix.

Corollary 3.9

If A is invertible, its determinant is non-zero.

$$\det(A) \neq 0$$

And

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

Corollary 3.10

If a matrix has two identical rows or columns, its determinant is zero.

$$\det(A) = 0$$

Lemma 3.11

Suppose that

$$\operatorname{rank}(A) = 0$$

Then

$$\det(A) = 0$$

3.5.4 Adjugate Matrix

Definition 3.4

The adjugate (or classical adjoint) of a square matrix A is the transpose of its cofactor matrix.

$$\operatorname{adj}(A) = \operatorname{Cof}(A)^T$$

$$\operatorname{adj}(A)_{ij} = C_{ij} = (-1)^{i+j} \det(A_{ij})$$

Theorem 3.12

For any square matrix A,

$$A \operatorname{adj}(A) = \operatorname{adj}(A) A = \det(A) I$$

Corollary 3.13

$$\operatorname{rank}(A) = 0 \implies \det(A) = 0 \implies \operatorname{adj}(A) = 0 \implies A \text{ is not invertible}$$

3.6 System of Linear System

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

- Are there any solutions?

- If there are solutions, how many are there?
- And how can we find the solutions if they exist?

3.6.1 Homogenous System

A homogenous system of linear equations is one in which all the constant terms are zero.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0 \end{cases}$$

Rank nullity theorem applies to homogenous systems,

$$\text{nullity}(A) = n - \text{rank}(A)$$

Therefore, the dimension of the solution space of a homogenous system is equal to the nullity of its coefficient matrix, so when the matrix is of full rank, the solution space is trivial, containing only the zero vector.

3.6.2 Non-Homogenous System

A non-homogenous system of linear equations is one in which at least one of the constant terms is non-zero.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

The solution set of a non-homogenous system can be described as the sum of a particular solution and the general solution of the corresponding homogenous system.

Lemma 3.14

Assume $x = u$ is a solution of

$$Ax = b$$

then the set of all solutions to this system is given by

$$\{u + v \mid v \text{ is a solution of } Ax = 0\}.$$

3.6.3 Solutions

Consider a system of linear equations

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

There is only three possibilities for the solution set of a system of linear equations,

1. No solution: The system is inconsistent and has no solution.
2. Unique solution: The system has exactly one solution.
3. Infinitely many solutions: The system has more than one solution, forming an affine subspace of dimension equal to the nullity of the coefficient matrix.

3.6.4 Augmented Matrix

The augmented matrix of a system of linear equations is obtained by appending the column of constants to the coefficient matrix.

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array}\right]$$

Notation

The augmented matrix of a system of linear equations $Ax = b$ is denoted by

$$[A \mid b].$$

And we can also denote the augmented matrix by

$$\hat{A}.$$

Lemma 3.15

The system $Ax = b$ has a solution if and only if the rank of the augmented matrix $\hat{A}$ is equal to the rank of the coefficient matrix A.

$$\operatorname{rank}(\hat{A}) = \operatorname{rank}(A)$$

3.6.5 Elementary Row Operations

Definition 3.5

Elementary row operations are operations that can be performed on the rows of a matrix without changing the solution set of the corresponding system of linear equations. There are three types of elementary row operations,

1. Row swapping: Interchanging two rows of the matrix.
2. Row scaling: Multiplying a row by a nonzero scalar.
3. Row addition: Adding a multiple of one row to another row.

The result of applying a sequence of elementary row operations to a matrix is called a row equivalent matrix. Two matrices are row equivalent if one can be obtained from the other by a sequence of elementary row operations.

Remark

The elementary row operations preserve kernel, rank and the solution set of the corresponding system of linear equations.

$$\begin{aligned} \ker(A) &= \ker(A'), \\ \operatorname{rank}(A) &= \operatorname{rank}(A'), \\ \operatorname{Sol}(Ax = b) &= \operatorname{Sol}(A'x = b) \end{aligned}$$

3.6.6 RREF

Definition 3.6

A matrix is in reduced row echelon form (RREF) if it satisfies the following conditions:

1. The leading entry in each nonzero row is 1.
2. Each leading 1 is the only nonzero entry in its column.
3. The leading 1 in each row is to the right of the leading 1 in the row above it.
4. Any rows consisting entirely of zeros are at the bottom of the matrix.

Remark

Every matrix is row equivalent to a unique matrix in reduced row echelon form.

Theorem 3.16

Using the elementary row operations, any matrix can be transformed into its unique reduced row echelon form.

Corollary 3.17

The rank of a matrix is the number of rows of REF that contain a leading 1.